

Project: Image Super-Resolution using Machine Learning

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Course: CSE 445

Section: 06

Group No: 04

Week 1 (Feb 5 – Feb 11): Understanding the Evaluation Approach

In the first week, I concentrated on understanding how the quality of reconstructed images would be evaluated in our super-resolution project. Since my role in the group involved testing and result analysis, I reviewed common evaluation methods used in image super-resolution research.

I studied the concept of Peak Signal-to-Noise Ratio (PSNR) and how it can be used to measure the similarity between reconstructed images and their original high-resolution versions. I also reviewed bicubic interpolation, which is commonly used as a baseline for comparison. Learning these concepts helped me prepare the evaluation strategy that would later be used to assess the performance of our machine learning model.

Week 2 (Feb 12 – Feb 19): Designing the Testing and Visualization Workflow

During the second week, while the team was completing the dataset preparation pipeline, I focused on setting up the testing and evaluation framework that would be used once we started experimenting with super-resolution models. At this stage, the team planned to test the ESPCN (Efficient Sub-Pixel Convolutional Neural Network) architecture for our image upscaling pipeline.

My responsibility was to prepare an evaluation process that would allow us to examine the performance of the ESPCN model. I studied how reconstructed images could be compared with their original high-resolution versions and explored image quality metrics such as Peak Signal-to-Noise Ratio (PSNR) for quantitative assessment.

In addition, I created a visualization format that could later be used to compare the results generated by different methods. The comparison layout included the low-resolution input image, bicubic interpolation output, model-generated image, and the original high-resolution ground truth image. Preparing this structure early helped us clearly evaluate the performance of the ESPCN model during the upcoming testing phase.

Week 3 (Feb 20 – Mar 1): Testing Pipeline Setup and ESPCN Evaluation

During this week, after the dataset preparation had been completed, I started implementing the testing pipeline for evaluating the super-resolution model. At that point in the project, our team was experimenting with the ESPCN (Efficient Sub-Pixel Convolutional Neural Network) architecture.

My role was to set up the evaluation process for the model outputs. I worked on loading the trained model and running inference on the test images to examine how effectively the model reconstructed high-resolution images from low-resolution inputs.

I also implemented the PSNR evaluation process, which allowed us to quantitatively measure the reconstruction quality of the generated images. Along with the numerical evaluation, I began preparing the visualization process to compare the ESPCN output with the bicubic interpolation baseline and the original ground truth images.

Week 4 (Mar 2 – Mar 8): Transition to SRCNN and Final Evaluation Setup

During this week, our team faced several technical difficulties with the ESPCN architecture, including unstable training and tensor shape alignment issues. After discussing these challenges, the team decided to move to the SRCNN (Super-Resolution Convolutional Neural Network) architecture as a more stable baseline model.

After this transition, I updated the testing pipeline so that it could work with the SRCNN model. I loaded the newly trained model weights and carried out evaluation on the test dataset. I calculated the PSNR values for the reconstructed images and compared them with the results obtained from bicubic interpolation.

I also completed the visualization component by generating comparison figures containing the low-resolution input, bicubic interpolation output, SRCNN reconstruction, and the original high-resolution image. These comparisons clearly showed the improvement achieved by the SRCNN model.